Periodic Hunter–Saxton Flow: Exact Two-Sided Lifespan and Universal Wave Breaking

Route-A source-local certificate HCS-C324

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Abstract

For every nonconstant periodic C^2 datum, we solve the once-integrated Hunter–Saxton equation throughout its maximal classical characteristic interval. One scalar factor gives the exact Jacobian, both lifespan endpoints, and every first-breaking label. We prove conservation of slope energy and the universal coefficient -2 at all simultaneous first breaking points, and close the constant-data and post-breaking boundaries. An asymmetric exact regression family separates the minimum-controlled future from the maximum-controlled past; a strict Route-A audit remains negative.

1 Exact characteristic theorem

Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and consider the once-integrated equation

$$u_{tx} + uu_{xx} + \frac{1}{2}u_x^2 = -\frac{1}{2}E, \qquad E = \int_{\mathbb{T}} u_x(t,x)^2\,dx. \tag{1}$$

This is our classical formulation for C^2 data; the differentiated third-order equation is not asserted pointwise at that regularity. Put $w_0 = u_{0x}$, $m = \min w_0$, and $M = \max w_0$. Nonconstant periodicity implies $E > 0$ and $m < 0 < M$.

Theorem 1 (Complete pre-breaking dynamics). *Define, for a characteristic label a,*

$$F(t,a) = \cos\frac{\sqrt{E}t}{2} + \frac{w_0(a)}{\sqrt{E}}\sin\frac{\sqrt{E}t}{2}, \tag{2}$$

$$T_+ = \frac{2}{\sqrt{E}}\arctan\frac{\sqrt{E}}{-m}, \qquad T_- = -\frac{2}{\sqrt{E}}\arctan\frac{\sqrt{E}}{M}. \tag{3}$$

There is a degree-one characteristic diffeomorphism $\eta(t,\cdot)$ for exactly $T_- < t < T_+$, normalized up to a common translation, such that

$$\eta_a = F^2, \qquad u_x(t,\eta(t,a)) = \frac{2F_t}{F} = \frac{-\sqrt{E}\sin(\sqrt{E}t/2) + w_0(a)\cos(\sqrt{E}t/2)}{F(t,a)}. \tag{4}$$

The first positive breaking labels are exactly $\operatorname{argmin} w_0$*; all of them break simultaneously. At each such label,*

$$u_x(t,\eta(t,a)) = -\frac{2}{T_+ - t} + O(1) \quad (t \uparrow T_+). \tag{5}$$

Backward degeneration occurs exactly at $\operatorname{argmax} w_0$ *and time T_-. No earlier degeneration occurs in either direction.*

Proof. Equation (1) yields along $\eta_t = u(t,\eta)$ the Riccati equation

$$\dot{w} = -\frac{1}{2}(w^2 + E), \qquad w(t,a) = u_x(t,\eta(t,a)).$$

Its solution is $w = 2F_t/F$. Since $\dot{\eta}_a = w\eta_a$ and $\eta_a(0,a) = 1$, this gives $\eta_a = F^2$. Moreover

$$\int_{\mathbb{T}} F^2\,da = \cos^2\frac{\sqrt{E}t}{2} + \sin^2\frac{\sqrt{E}t}{2} = 1,$$

because $\int w_0 = 0$ and $\int w_0^2 = E$. Conversely this identity makes the construction explicit: set

$$\eta(t,a) = b(t) + \int_0^a F(t,s)^2\,ds, \qquad u(t,\eta(t,a)) = \eta_t(t,a),$$

where $b(0) = 0$ and $b'(0) = u_0(0)$. Then $\eta(0,a) = a$, $u(0,a) = u_0(a)$, and differentiating in a gives $u_x \circ \eta = \eta_{ta}/\eta_a = 2F_t/F$. Its Riccati equation is precisely (1) along every characteristic. The free common translation $b(t)$ is the harmless additive gauge. Thus the construction is a degree-one circle diffeomorphism exactly while F is positive.

For fixed $t > 0$ before the first pole, F is increasing in w_0. Its first zero is therefore attained precisely where $w_0 = m$, and solving $F = 0$ gives T_+. Reversing time makes the maximum decisive and gives T_-. At a future first label, $F(T_+, a) = 0$ and $F_t(T_+, a) \neq 0$. Thus $2F_t/F = 2/(t - T_+) + O(1)$, proving (5) and the complete label statement. □

2 Energy, degeneracies, and the endpoint

The same representation closes the conservation law without dividing by the Jacobian at its endpoint:

$$w(t,a)^2\eta_a(t,a) = E\left[-\sin\frac{\sqrt{E}t}{2} + \frac{w_0(a)}{\sqrt{E}}\cos\frac{\sqrt{E}t}{2}\right]^2. \tag{6}$$

Integrating (6), the mixed term vanishes and the remaining two terms sum to E. Consequently $\int_{\mathbb{T}} u_x^2\,dx = E$ throughout (T_-, T_+), consistently closing (1).

If $E = 0$, then $w_0 = 0$ and the solution is spatially constant; neither formula in (3) is interpreted by division through zero. A flat interval or several isolated points may realize the global minimum, and Theorem 1 includes the entire minimizing set rather than a generic singleton. At $t = T_+$ the map ceases to be a diffeomorphism. This paper neither constructs nor selects a conservative, dissipative, or other weak continuation.

3 Asymmetric audit and Route-A boundary

Single harmonics cannot detect an accidental exchange of m and M. For $\theta = 2\pi k a$, take

$$w_0 = \cos\theta + \tfrac{1}{2}\cos 2\theta = (\cos\theta + \tfrac{1}{2})^2 - \tfrac{3}{4}.$$

Then $E = 5/8$, $M = 3/2$ occurs at k labels, and $m = -3/4$ occurs at $2k$ labels. Its negative swaps the extrema and multiplicities. The certificate checks both signs for $k = 1, 2, 3$, in addition to twelve Pythagorean single harmonics. It records 2,354 scalar leaves. The producer-independent checker performs 3,857 checks, SymPy closes 1,508 identities, two isolated replays are byte exact, and 60 hostile mutations are rejected. These are convention regressions; the proof above covers all declared data.

The nearest registered models have different mechanisms: C195 is viscous Burgers smoothing, C256 is dispersive KdV cnoidal motion, and C278 is a finite-dimensional Camassa–Holm two-peakon weak manifold.

Under `NO_BAD_EULER_OR_ROOT_NUMBER`, the strict tuple is

$$(\text{A0\_FAIL}, \text{A1\_FAIL}, \text{A2\_FAIL}, \text{A3\_FAIL}, \text{A4\_FORMAL\_HINT}).$$

The geometric interpretation is only a formal hint. There is no arithmetic prime owner, primitive-orbit Euler ledger, Euler product, target functional equation, counting law, target-zero match, or Hilbert–Pólya operator. Route A is rejected and Route B stays locked. No target local data, root number, or automorphy is claimed.

AI use. A generative language model assisted drafting and code scaffolding. The displayed proof, independent recomputation, hostile tests, and deterministic artifacts define the audit record.

Source lineage

References

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